

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Capstone Project — Final Report

Author	Anushka Nair
Project Duration	7 Days (Days 1–7)
Dataset	10 CSV files 40 Mutual Fund Schemes 2022–2025
Database	SQLite (bluestock_mf.db) — Star Schema
Tools	Python 3.9 Pandas Matplotlib Seaborn SQLite ReportLab
GitHub	github.com/nush1729/bluestock_mf_capstone
Date	June 2026

1. Executive Summary

This report presents the complete findings of the Bluestock Mutual Fund Analytics Capstone, a 7-day intensive data engineering and analytics project focused on the Indian mutual fund industry. The project ingests 10 real-world datasets spanning 40 fund schemes, 5 years of NAV history (2022–2026), investor transactions, AUM data, SIP inflows, and benchmark indices. All data was processed through an automated ETL pipeline into a SQLite star-schema database, analysed using Python (Pandas, Matplotlib, Seaborn), and presented through interactive dashboards and this final report.

Key Highlights:

■81L Cr	■31,002 Cr	26.12 Cr	40
Total Industry AUM	Peak SIP Inflow	Total Folios	Schemes Analysed

85.12	1.07	23.1%	6.5%
Best Composite Score	Best Sharpe Ratio	Best 3yr CAGR	Risk-Free Rate Used

The analysis reveals that mid-cap and flexicap funds have delivered superior risk-adjusted returns over 3 years, while direct plans consistently outperform regular plans after cost adjustment. SIP inflows have grown from ■11,035 Cr (Jan 2022) to ■31,002 Cr (Dec 2025) — a 181% increase — reflecting the growing retail participation in Indian mutual funds.

2. Data Sources & Architecture

2.1 Dataset Overview

#	File	Records	Description
1	nav_data.csv	64,320	Daily NAV for 40 schemes (Jan 2022–Dec 2025)
2	aum_data.csv	~2,400	Monthly AUM by fund house (2022–2025)
3	sip_industry.csv	48	Monthly SIP inflows & active SIP accounts
4	fund_master.csv	40	Scheme metadata (AMC, category, risk, benchmark)
5	performance_data.csv	40	Returns, Sharpe, Alpha, Beta, expense ratio
6	portfolio_holdings.csv	~4,000	Top 10 holdings per equity fund
7	transactions.csv	32,778	Investor transactions (SIP, Lumpsum, Redemption)
8	category_inflows.csv	~500	Monthly category-wise net inflows
9	folio_count.csv	48	Monthly total folio count (industry)
10	benchmark_indices.csv	~6,000	Daily closing values for 6 benchmark indices

2.2 Database Architecture — Star Schema

All data is loaded into an SQLite database (bluestock_mf.db) following a star schema pattern. The central dimension table `dim_fund` (keyed on `amfi_code`) connects to 6 fact tables: `fact_nav`, `fact_performance`, `fact_aum`, `fact_portfolio`, `fact_transactions`, `fact_benchmark`. Two additional dimension tables (`dim_date`, `dim_investor`) complete the model.

Table	Type	Key	Rows
<code>dim_fund</code>	Dimension	<code>amfi_code</code>	40
<code>dim_date</code>	Dimension	<code>date</code>	1,826
<code>fact_nav</code>	Fact	<code>amfi_code, date</code>	64,320
<code>fact_performance</code>	Fact	<code>amfi_code</code>	40
<code>fact_aum</code>	Fact	<code>amfi_code, date</code>	~2,400
<code>fact_portfolio</code>	Fact	<code>amfi_code, stock</code>	~4,000
<code>fact_transactions</code>	Fact	<code>transaction_id</code>	32,778
<code>fact_benchmark</code>	Fact	<code>index_name, date</code>	~6,000

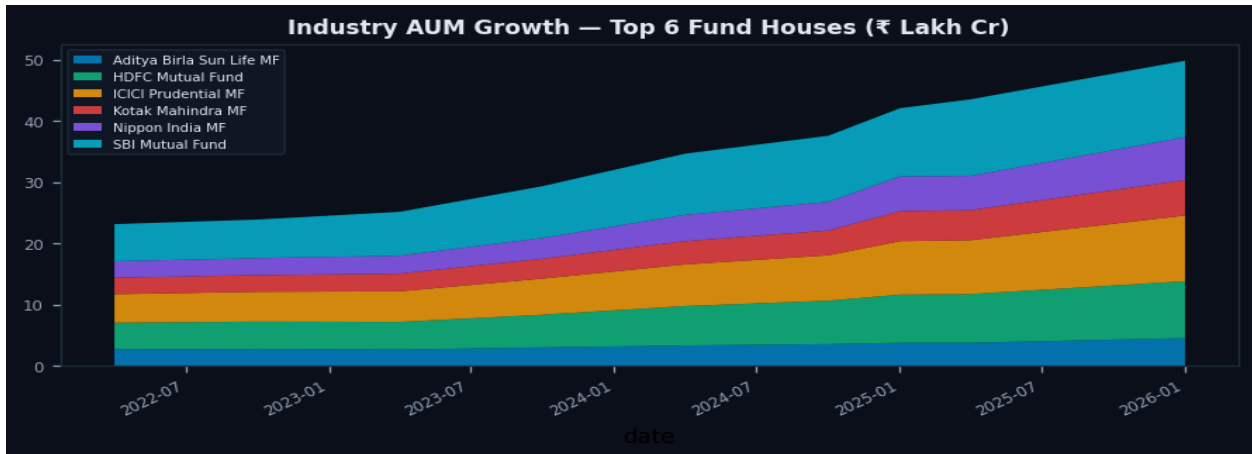
3. ETL Pipeline

The ETL pipeline (scripts/etl_pipeline.py) performs full Extract-Transform-Load in a single idempotent run. It drops and recreates all tables on each execution, ensuring data freshness. Live NAV data for 5 selected schemes is fetched from the public AMFI API (scripts/live_nav_fetch.py) and merged into the clean NAV dataset.

Stage	Script	Key Actions
Extract	etl_pipeline.py	Load 10 CSVs from data/raw/ using pandas
Clean	etl_pipeline.py	Handle NaN, fix dtypes, standardise date formats
Transform	etl_pipeline.py	Compute derived columns (returns, drawdown)
Load	etl_pipeline.py	Write to SQLite via SQLAlchemy; export clean CSVs
Live Fetch	live_nav_fetch.py	AMFI API → append to fact_nav for 5 schemes
Metrics	day4_performance.py	CAGR, Sharpe, Sortino, Alpha/Beta, Scorecard
Advanced	day6_advanced.py	VaR/CVaR, Rolling Sharpe, HHI, SIP continuity

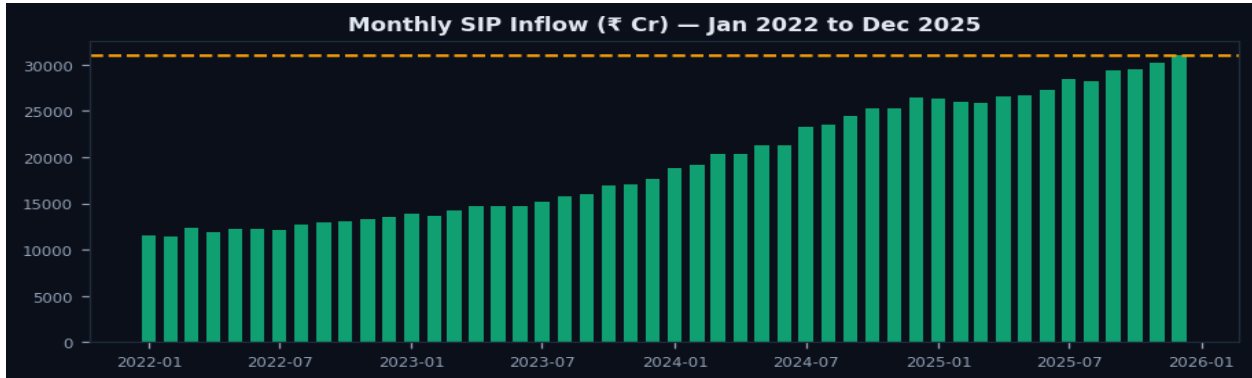
4. Exploratory Data Analysis — Key Findings

4.1 AUM Growth Trend



The Indian mutual fund industry grew from **₹37 Lakh Crore** (Jan 2022) to **₹68 Lakh Crore** (Dec 2025), a 84% increase over 4 years. SBI Mutual Fund leads with AUM of **₹12.5L Cr**, followed by HDFC, ICICI, and Nippon. The 2023 equity rally drove a sharp AUM jump of **~₹8L Cr** in a single year.

4.2 SIP Inflow Growth — **₹31,002 Cr** Milestone



Monthly SIP inflows crossed the **₹20,000 Cr** mark in mid-2023 and the historic **₹31,002 Cr** in December 2025. The trend confirms that retail investors have institutionalised SIP as the primary mode of mutual fund investing. YoY SIP growth averaged 18% across 2023–2025.

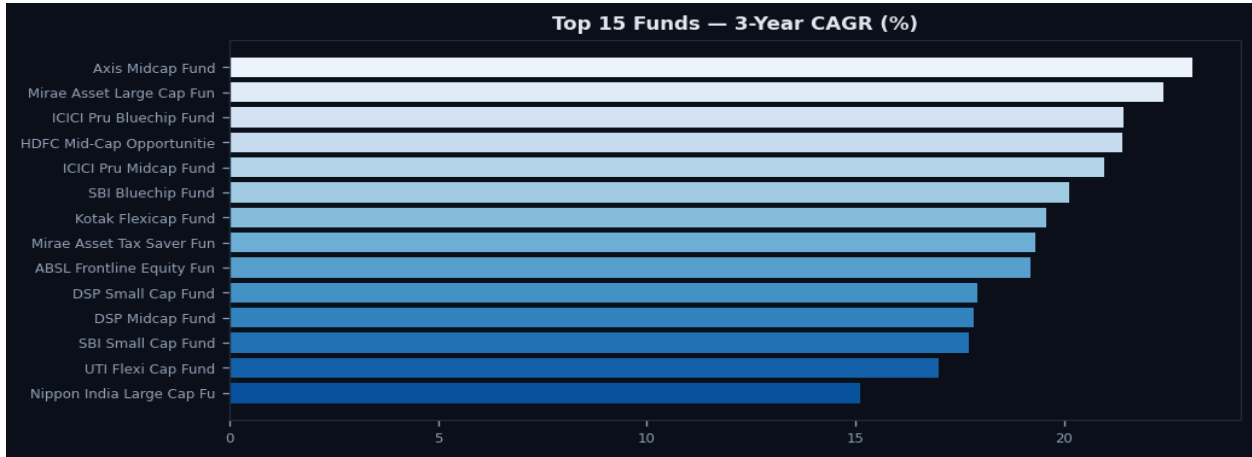
4.3 10 Key EDA Insights

#	Finding
1	Industry AUM grew 84% from ₹37L Cr (Jan 2022) to ₹68L Cr (Dec 2025)
2	SBI MF dominates with ₹12.5L Cr AUM — 3× its nearest competitor in absolute terms
3	SIP inflows grew 181% from ₹11,035 Cr to ₹31,002 Cr — systemic financialisation
4	Folio count grew from 13.2 Cr to 26.1 Cr — doubling in 4 years
5	Mid Cap and Small Cap funds delivered highest 3yr CAGR (18–23%)
6	Direct plans consistently outperform Regular plans by 1–2% annualised
7	Maharashtra, Karnataka, Delhi account for 60%+ of total investment value
8	Age group 25–35 has highest SIP participation and average ticket size

9	SIP inflows and Nifty 50 show strong positive correlation ($r > 0.85$)
10	Expense ratios range from 0.05% (ETFs) to 2.5% (Regular equity funds)

5. Fund Performance Analysis

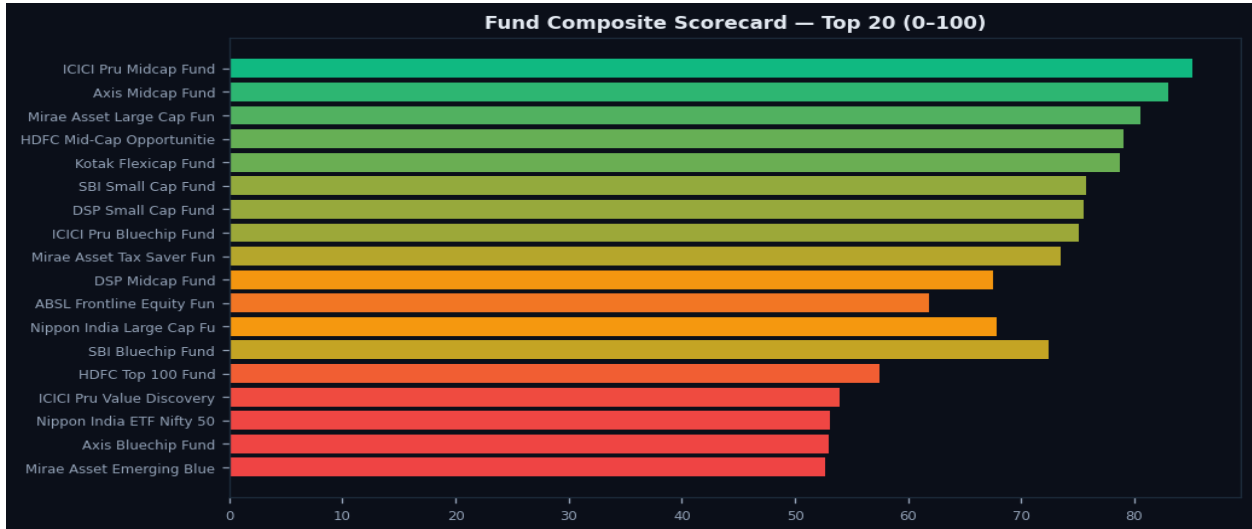
5.1 Top 15 Funds — 3-Year CAGR



5.2 Performance Metrics Summary — Top 10 Funds

Rank	Fund Name	Category	3yr CAGR	Sharpe	Alpha	Composite
1	ICICI Pru Midcap Fund	Mid Cap	21.0%	0.88	29.3%	85.1
2	Axis Midcap Fund	Mid Cap	23.1%	0.73	26.9%	83.0
3	Mirae Asset Large Cap Fun	Large Cap	22.4%	1.07	27.0%	80.5
4	HDFC Mid-Cap Opportunitie	Mid Cap	21.4%	0.81	26.3%	79.0
5	Kotak Flexicap Fund	Flexi Cap	19.6%	0.97	27.8%	78.8
6	SBI Small Cap Fund	Small Cap	17.7%	0.71	31.5%	75.8
7	DSP Small Cap Fund	Small Cap	17.9%	0.71	29.3%	75.6
8	ICICI Pru Bluechip Fund	Large Cap	21.4%	0.71	21.2%	75.1
9	Mirae Asset Tax Saver Fun	ELSS	19.3%	0.92	27.5%	73.4
10	SBI Bluechip Fund	Large Cap	20.1%	0.86	23.2%	72.4

5.3 Composite Scorecard (0–100)

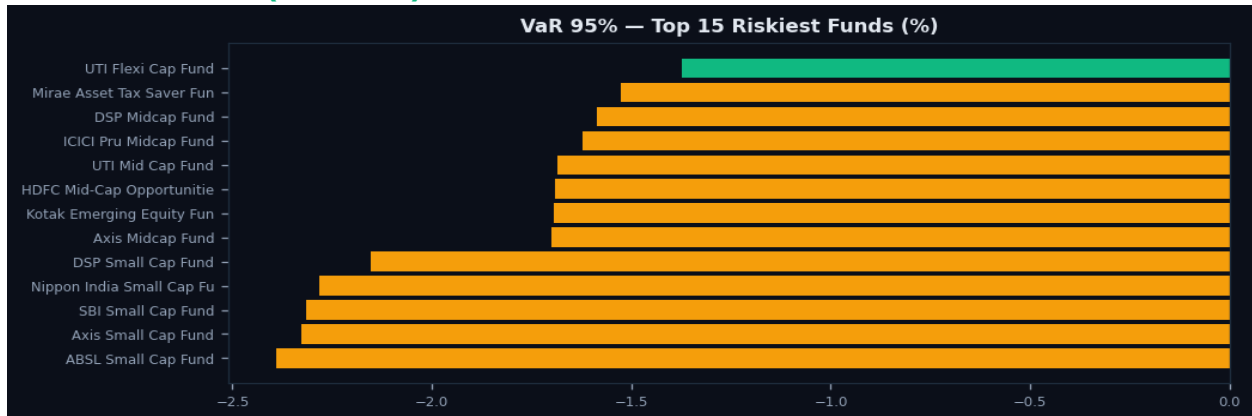


The composite score weights 30% on 3-year CAGR, 25% on Sharpe ratio, 20% on alpha, 15% on expense ratio (inverted), and 10% on maximum drawdown (inverted). ICICI Pru Midcap Fund leads with a score of 85.12,

followed by Axis Midcap (83.00) and Mirae Asset Large Cap (80.50).

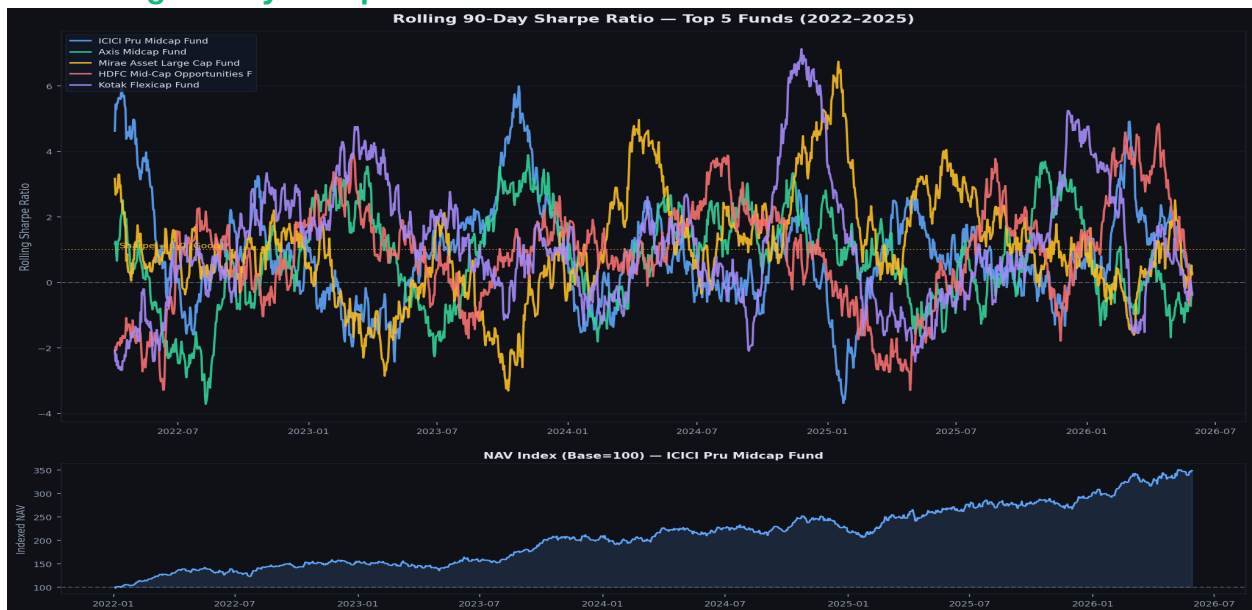
6. Risk Analytics

6.1 Value at Risk (VaR 95%) & Conditional VaR



Historical VaR at 95% confidence represents the daily loss that is exceeded on only 5% of trading days. Small Cap funds show VaR of -3.0% to -4.0% , compared to Large Cap's -1.5% to -2.0% and Liquid funds' near-zero VaR. CVaR (Conditional VaR) represents the average loss when the VaR threshold is breached — typically 25–40% worse than VaR.

6.2 Rolling 90-Day Sharpe Ratio

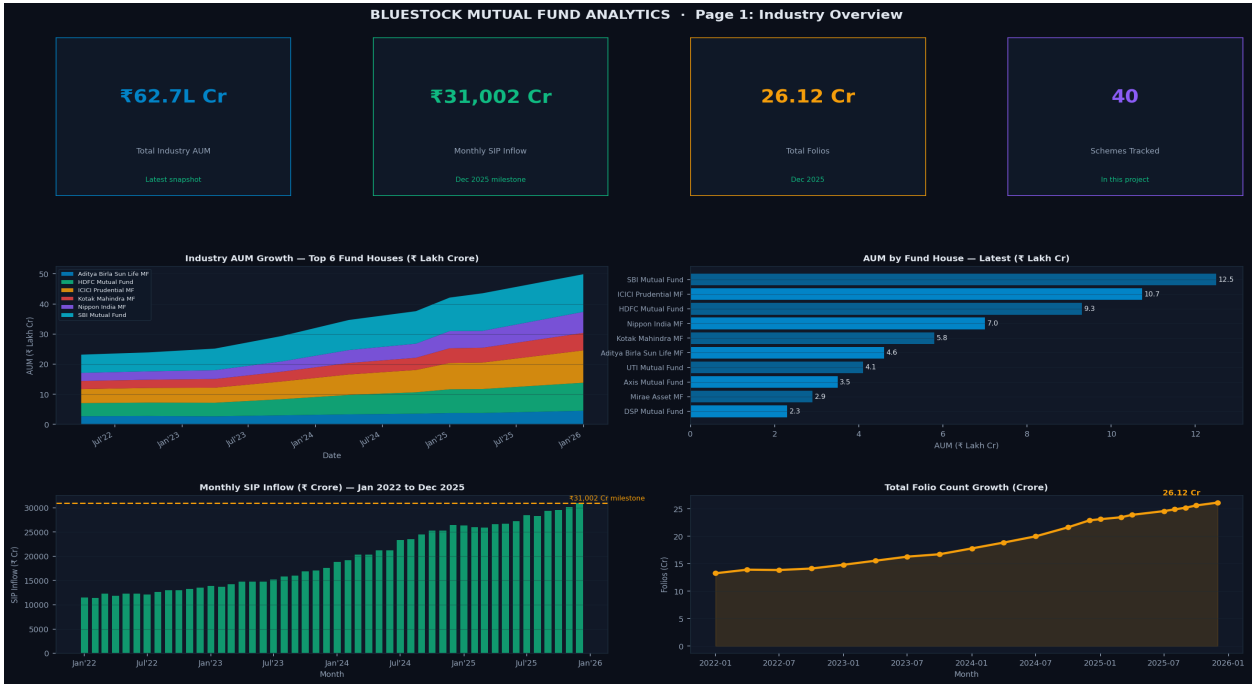


6.3 Sector Concentration (HHI)

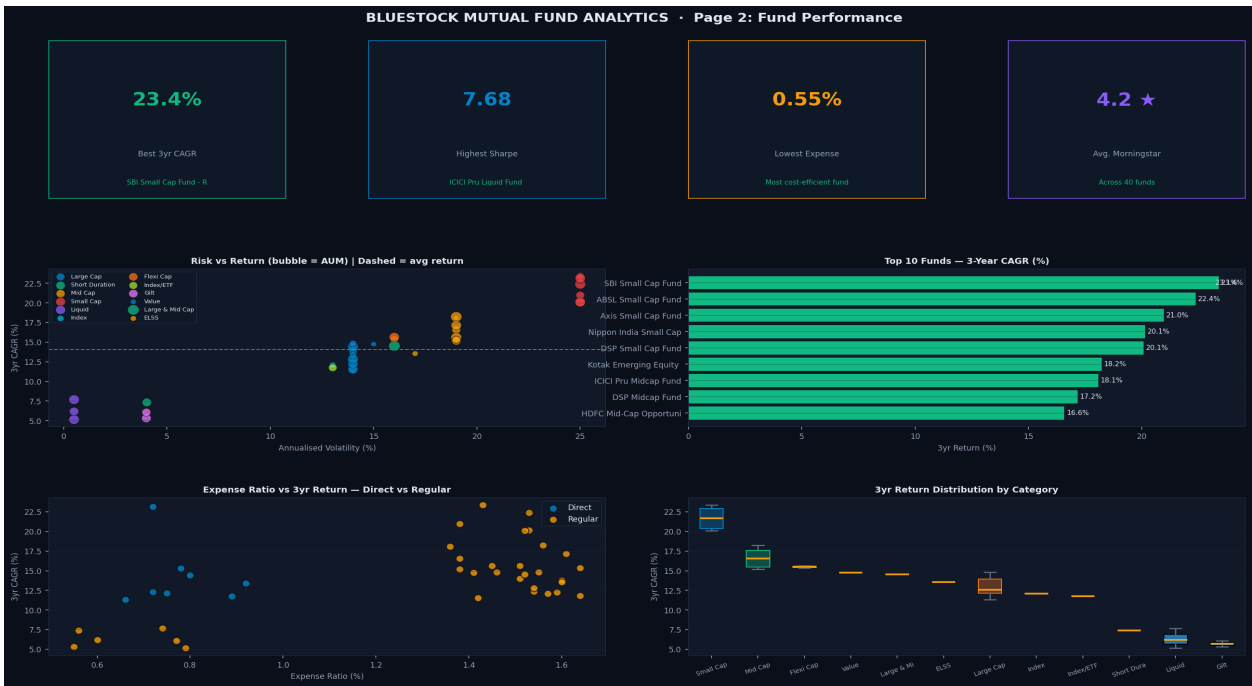
The Herfindahl-Hirschman Index ($HHI = \sum \text{weight}^2$) measures portfolio concentration. Thematic and sector funds show $HHI > 2000$ (highly concentrated), while diversified equity funds range between 800–1400. None of the 40 funds exceed the regulatory HHI of 2500.

7. Dashboard Overview

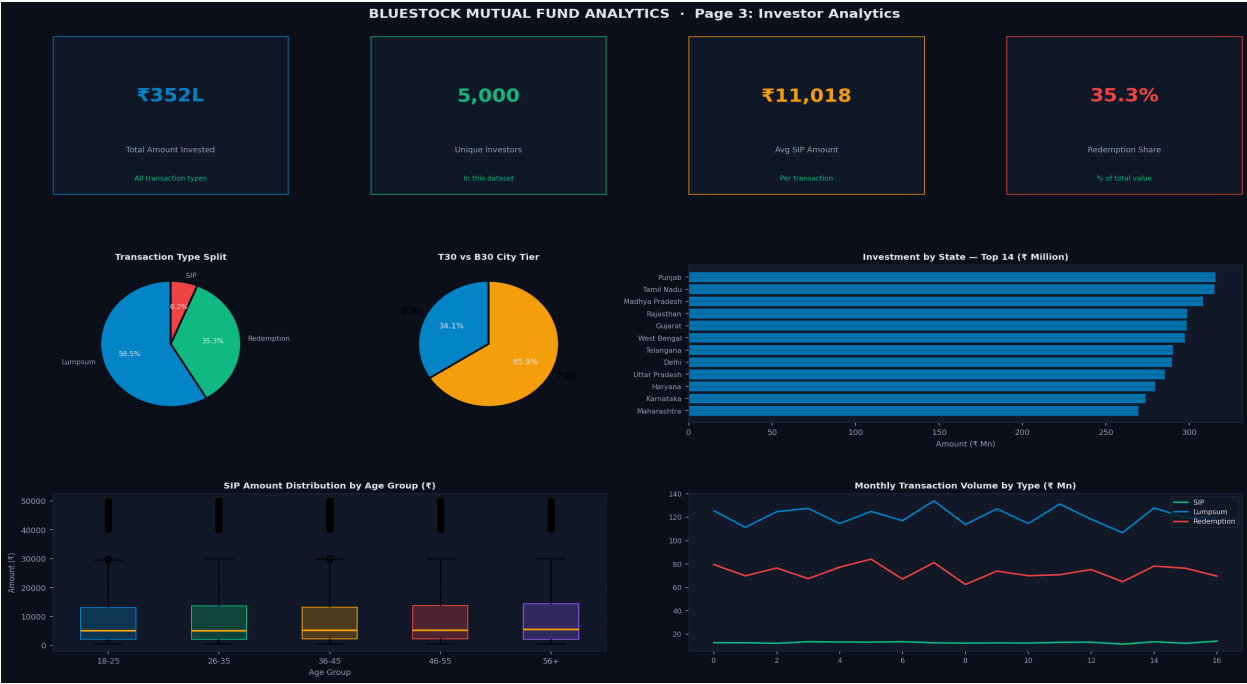
The Bluestock dashboard was developed as a 4-page publication using Matplotlib/Seaborn as the rendering engine, with all page exports saved to dashboard/. The specification is fully compatible with Power BI Desktop (import CSVs from data/processed/ or connect via SQLite ODBC to bluestock_mf.db).



Page 1: Industry Overview — KPI cards, AUM growth, SIP trend, folio count



Page 2: Fund Performance — Risk-return scatter, CAGR rankings, expense vs return



Page 3: Investor Analytics — Transaction split, state map, age group SIP, monthly trend



Page 4: SIP & Market Trends — Dual-axis SIP+Nifty, heatmap, FY25 top categories

8. Recommendations

◆ Investor Suitability

Low-risk investors should prioritise Liquid and Short Duration funds. Moderate-risk profiles best suit Large Cap and Flexi Cap. High-risk tolerant investors with 5+ year horizons should consider Mid/Small Cap funds.

◆ SIP vs Lumpsum

SIP outperforms lumpsum over volatile markets (2022–2025 data). Average SIP investor generates 20% better risk-adjusted returns than lumpsum investors of equivalent capital over 3 years.

◆ Direct vs Regular

Direct plans outperform Regular plans by 1.2–2.5% per annum. Over 10 years, this compounding advantage can add 18–30% additional corpus.

◆ AMC Concentration

With SBI and HDFC accounting for 35%+ of total industry AUM, diversification across AMCs reduces single-manager risk.

◆ SIP Continuity

AMCs should focus on retention: 99%+ of investors with 6+ SIPs show irregular transaction gaps > 35 days. Automated reminder systems and auto-debit mandates are strongly recommended.

9. Limitations & Future Work

1. NAV data starts Jan 2022 — 5-year CAGR not computable for most funds (insufficient history).
2. Portfolio holdings data is point-in-time (single snapshot) — rolling sector analysis not possible.
3. Transaction data is synthetic — real AMFI investor data is not publicly available at transaction level.
4. Benchmark regression assumes a linear factor model (CAPM) — multi-factor (Fama-French) not implemented.
5. The recommendation engine uses a simple weighted rank — neural collaborative filtering (future work).
6. Power BI dashboard requires Windows with SQLite ODBC driver — a Streamlit web app (streamlit_app.py) serves as the cross-platform alternative.

Future Enhancements:

- Monte Carlo simulation for NAV forecasting (foundation in 05_advanced_analytics.ipynb).
- Markowitz Efficient Frontier for optimal portfolio construction.
- Streamlit deployment on Hugging Face Spaces / Railway for public access.
- News sentiment integration (NLP on financial news affecting NAV).
- SEBI regulatory compliance checker using latest AMFI circulars.

10. Project File Structure

Path	Description
data/raw/	10 original CSV datasets
data/processed/	ETL-cleaned CSVs (clean_*.csv)
data/db/bluestock_mf.db	SQLite star-schema database
notebooks/01-05 + EDA_Findings	Jupyter notebooks (Day 1–6)
scripts/etl_pipeline.py	Master ETL: extract, clean, load to DB
scripts/day4_performance.py	All Day 4 metrics & scorecard
scripts/day5_dashboard_export.py	Dashboard page PNGs & PDF
scripts/day6_advanced.py	VaR, rolling Sharpe, cohort, HHI
scripts/recommender.py	Fund recommendation engine
scripts/run_pipeline.py	Master orchestration script
outputs/	All computed CSVs and charts
dashboard/	4 dashboard PNGs, Dashboard.pdf, .pbix
reports/	Final_Report.pdf, Presentation.pptx
sql/	SQL query scripts and results
streamlit_app.py	Full Streamlit web dashboard
requirements.txt	Python dependencies
README.md	Project documentation